

# Dr. Eduard Keilmann

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## RESEARCH INTERESTS

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My research focuses on molecular-cloud formation and star formation across a broad range of spatial scales and using diverse observational tracers, from dust continuum emission to atomic and molecular spectral lines. I am particularly interested in the dynamics of the interstellar medium (ISM), giant molecular-cloud properties, cloud assembly and dispersal, and the role of stellar feedback in regulating star formation in the Milky Way and nearby galaxies.

## ACADEMIC & RESEARCH APPOINTMENTS

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- 2025–present      **UNIVERSITY OF COLOGNE** | *Cologne, Germany*  
**Postdoctoral Researcher in Astrophysics**  
Investigate the formation, structure, evolution, and dispersal of molecular clouds and star-forming regions, with emphasis on stellar feedback and gas dynamics in the Milky Way and nearby galaxies. Analyze spectrally resolved far-infrared, submillimeter, and radio data, including SOFIA/upGREAT [C II] 158  $\mu\text{m}$  and [O I] 63  $\mu\text{m}$  observations, Herschel dust-continuum maps, CO spectroscopy, and H I data. Develop Python-based analysis and modeling pipelines for large astronomical data sets and spectral cubes (100 GB+; millions of spectra), including column-density mapping, cloud identification, spectral analysis, clustering, PCA, and other statistical/ML methods. Apply radiative-transfer and photodissociation-region (PDR) modeling to infer physical conditions and interpret atomic and molecular line emission. Author/co-author of 10 refereed publications, including 4 first-author papers.
- 04/2021–06/2025      **UNIVERSITY OF COLOGNE** | *Cologne, Germany*  
**Doctoral Researcher in Astrophysics**  
Doctoral research on molecular-cloud formation, star formation, and stellar feedback from Galactic clouds to giant molecular clouds in M33.

## EDUCATION

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- 2021–2025      **PhD in Astrophysics** | University of Cologne | *Cologne, Germany*  
Dissertation: “The Genesis of Stars: From Giant Molecular Clouds to Star-Forming Cores”  
Advisors: Prof. Dr. Jürgen Stutzki and Dr. Nicola Schneider | Grade: 1.0 (magna cum laude)
- 2016–2019      **M.Sc. in Physics** | Johannes-Gutenberg University Mainz | *Mainz, Germany*  
Thesis: “Standard Model Effective Field Theory Effects in Dijet Events at Tevatron” | Grade: 1.3  
Focus: Elementary Particle Physics and Quantum Field Theory | Minor: Meteorology (Atmospheric Hydrodynamics)
- 2011–2016      **B.Sc. in Physics** | Johannes-Gutenberg University Mainz | *Mainz, Germany*  
Thesis: “Massenbestimmung steriler Neutrinos anhand von Supernovae” | Minors: Mathematics and Computer Science

## FELLOWSHIPS & AWARDS

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### Walter Benjamin Fellowship — German Research Foundation (DFG)

(awarded 2026). Two-year independent postdoctoral research fellowship for a research stay at NASA Jet Propulsion Laboratory (JPL), California Institute of Technology.

## SCIENTIFIC COLLABORATIONS & AFFILIATIONS

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- SFB 1601 — “Habitats of Massive Stars across Cosmic Time”** — Member of the University of Cologne Collaborative Research Center, Project B2; studies massive-star feedback and its gaseous environments using velocity-resolved atomic and molecular spectroscopy.
- NASA ASTHROS** — Member of the science team for the Astrophysics Stratospheric Telescope for High Spectral Resolution Observations at Submillimeter-wavelengths, a high-altitude balloon mission with a major science focus on stellar feedback.
- SOFIA FEEDBACK Consortium** — Member of the legacy survey studying stellar feedback in massive star-forming regions through velocity-resolved [C II] 158  $\mu\text{m}$  and [O I] 63  $\mu\text{m}$  observations with SOFIA/upGREAT.

## RESEARCH & TECHNICAL EXPERTISE

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**Observational astrophysics:** Far-infrared/submillimeter/radio spectroscopy and continuum analysis; ISM, molecular clouds, star formation, stellar feedback, PDRs; SOFIA/upGREAT, Herschel, CO and H I data.

**Scientific data analysis:** Spectral-cube processing; line-profile and self-absorption analysis; SED/column-density modeling; N-PDFs; cloud segmentation/dendrograms; statistical characterization of GMCs.

**Scientific computing:** Python, R, SQL, Bash; NumPy, SciPy, pandas, Astropy, Matplotlib, scikit-learn, PyTorch, CLASS/GILDAS, Git.

**Data science & machine learning:** PCA, clustering/Gaussian mixture models, regression, Random Forests, surrogate models, signal/image processing, image segmentation, neural-network prototyping.

**Software engineering:** C++, Java, C#, .NET/WPF, MVVM; modular architecture, unit testing, version control, CI/CD.

**Languages:** German (native), English (fluent), Norwegian (beginner).

## TEACHING EXPERIENCE

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2021–2025 **UNIVERSITY OF COLOGNE** | *Cologne, Germany*

### Teaching in Physics / Astrophysics

Led student tutorial sessions, designed problem sets and supporting materials, and graded assignments and examinations.

## ADDITIONAL PROFESSIONAL EXPERIENCE

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10/2019–03/2021 **HDI** | *Cologne, Germany*

### Data Scientist (Actuary)

Developed and automated Solvency II life-insurance models in R; integrated and validated heterogeneous data sources and performed model validation, analysis, and regulatory reporting.

06/2019–09/2019 **HOTTGENROTH** | *Cologne, Germany*

### Software Developer

Developed modular energy-simulation software in C#/.NET/WPF, integrating climate data and physics-based building parameters.

06/2017–12/2018 **BIONTECH** | *Mainz, Germany*

### Student Software Developer / Software Architecture

Contributed to modular .NET/C# software systems and architecture for biomedical research applications, including collaboration with U.S.-based teams.

09/2014–12/2015 **JOHANNES-GUTENBERG UNIVERSITY MAINZ** | *Mainz, Germany*

### Research Assistant — Quantum Physics

Built and tested acousto-optic modulator driver systems for high-precision laser experiments; performed signal testing and iterative hardware/software debugging.

09/2006–06/2009 **DEUTSCHE TELEKOM** | *Mainz, Germany*

### IT Specialist (Fachinformatiker), Apprenticeship

Developed software and data-processing tools in C++ and related technologies.

## PROFESSIONAL DEVELOPMENT

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**11/2025** “Foundations and Applications of Machine Learning” — Max Planck Institute for Radio Astronomy (MPIfR), Bonn

**09/2024** “Introduction to Machine Learning and PyTorch” — University of Cologne

## SUCCESSFUL OBSERVING PROPOSALS

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**Principal Investigator:** APEX telescope time in 2022 and 2024 (36.3 h total) for [C I] observations in M33 and studies of interstellar bubbles.

**Co-Investigator:** IRAM 30 m and NOEMA telescope time in 2024–2025 (42.8 h total) for molecular-gas, shock, and feedback studies in Cygnus X.

## SELECTED SCIENTIFIC PRESENTATIONS

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**09/2023** **German Astronomical Society, Berlin** — contributed talk: FEEDBACK observations of RCW79

**04/2024** **Heritage of SOFIA — Scientific Highlights and Future Perspectives, Stuttgart** — poster: First detection of the [C II] 158  $\mu\text{m}$  line in the intermediate-velocity cloud Draco

**07/2024** **Research Group Prof. Walch-Gassner, University of Cologne** — invited seminar: M33 molecular-cloud matching

**09/2024** **German Astronomical Society, Cologne** — contributed talk: M33 molecular-cloud matching

**05/2025** **Star Formation, Stellar Feedback, and the Ecology of Galaxies, Visegrád** — poster: Reassessing the [C II]-deficit in RCW79

## PUBLICATION LIST

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**10 refereed publications, including 4 first-author papers.** My name is shown in bold; publication titles link to the journal DOI.

1. Bonne, L., Schneider, N., Dannhauer, S., **Keilmann, E.**, Jackson, J. M., Simon, R., et al. (2026). "[Molecular cloud dispersal traced by the ionized carbon 158  \$\mu\text{m}\$  line](#)" *Astronomy & Astrophysics*, 711, A304.
2. Schneider, N., Dannhauer, S., **Keilmann, E.**, Kabanovic, S., Topkaras, T., Ossenkopf-Okada, V., et al. (2025). "[Globules and pillars in Cygnus X. IV. Velocity-resolved \[O I\] 63  \$\mu\text{m}\$  map of a peculiar proplyd-like object](#)" *Astronomy & Astrophysics*, 704, A271.
3. Dannhauer, S. M., Vider, S., Schneider, N., Simon, R., Comerón, F., **Keilmann, E.**, et al. (2025). "[The Diamond Ring in Cygnus X: Advanced stage of an expanding bubble of ionised carbon](#)" *Astronomy & Astrophysics*, 703, A197.
4. Schneider, N., Ossenkopf-Okada, V., Röllig, M., Seifried, D., Klessen, R. S., Kritsuk, A. G., **Keilmann, E.**, et al. (2025). "[The H I-to-H<sub>2</sub> transition in the Draco cloud](#)" *Astronomy & Astrophysics*, 699, A354.
5. **Keilmann, E.**, Dannhauer, S., Kabanovic, S., Schneider, N., Ossenkopf-Okada, V., Simon, R., et al. (2025). "[\[C II\]-deficit caused by self-absorption in an ionized carbon-filled bubble in RCW79](#)" *Astronomy & Astrophysics*, 697, L2.
6. **Keilmann, E.**, Kabanovic, S., Schneider, N., Ossenkopf-Okada, V., Stutzki, J., Kobayashi, M. I. N., et al. (2024). "[Molecular cloud matching in CO and dust in M33. II. Physical properties of giant molecular clouds](#)" *Astronomy & Astrophysics*, 692, A226.
7. **Keilmann, E.**, Buchbender, C., Ossenkopf-Okada, V., Schneider, N., Kabanovic, S., Stutzki, J., et al. (2024). "[Molecular cloud matching in CO and dust in M33. I. High-resolution hydrogen column density maps from Herschel](#)" *Astronomy & Astrophysics*, 688, A171.
8. Schneider, N., Ossenkopf-Okada, V., **Keilmann, E.**, Röllig, M., Kabanovic, S., Bonne, L., et al. (2024). "[First detection of the \[C II\] 158  \$\mu\text{m}\$  line in the intermediate-velocity cloud Draco](#)" *Astronomy & Astrophysics*, 686, A109.
9. Bonne, L., Kabanovic, S., Schneider, N., Zavagno, A., **Keilmann, E.**, Simon, R., et al. (2023). "[The SOFIA FEEDBACK \[C II\] Legacy Survey: Rapid molecular cloud dispersal in RCW 79](#)" *Astronomy & Astrophysics*, 679, L5.
10. **Keilmann, E.**, & Shepherd, W. (2019). "[Dijets at Tevatron cannot constrain SMEFT four-quark operators](#)" *Journal of High Energy Physics*, 2019(9), 86.